

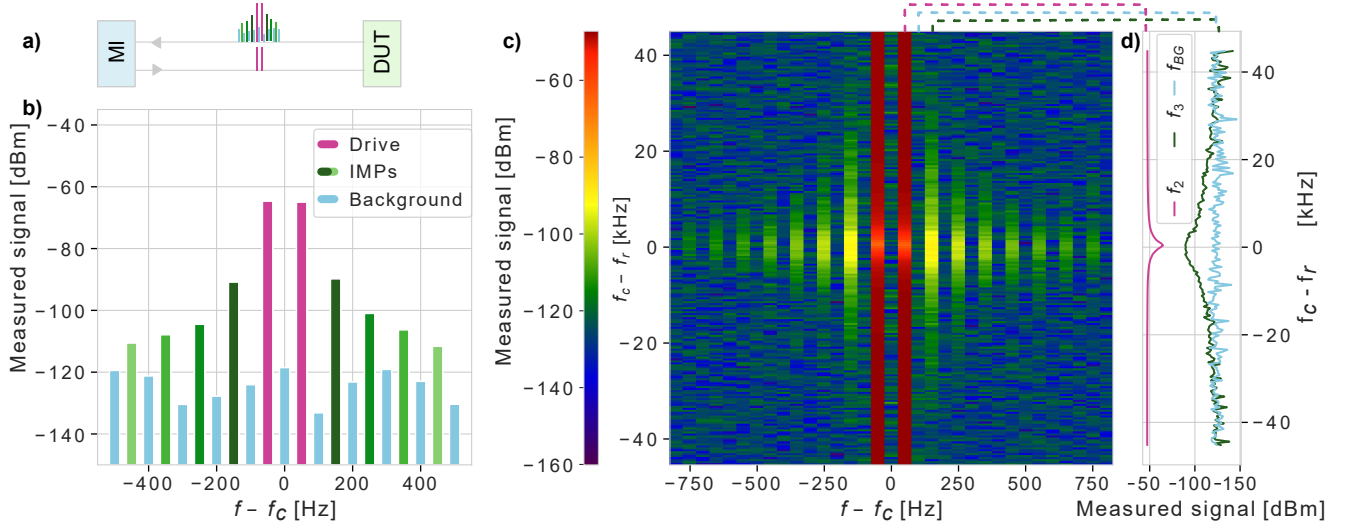


FIG. 1. Intermodulation measurements at drive powers corresponding to $\langle n \rangle = 10^3$ when on resonance. **a)** A schematic of the tones generated at the output of the measurement instrumentation (MI) and the output of the device-under-test (DUT). **b)** IMPs measured when two drive tones are applied at resonance. The abscissa shows the detection comb frequencies f relative to the centre of the detection comb $f_c = (f_1 + f_2)/2$. IMPs are detected at frequencies fulfilling the $f_{\text{IMP}} = k_1 f_1 + k_2 f_2$ condition, which falls on every other tone in this detection comb. **c)** IMP power spectroscopy performed by sweeping the frequency comb depicted in a) across a frequency spectrum centered around the resonance frequency f_r . **d)** Vertical line-cuts of the IMP spectroscopy data shown in c): the device resonance demonstrates as a dip in the transmission of drive tone f_1 , the third IMP f_3 presents as a peak at $2f_2 - f_1$, and the background noise is shown at a frequency f_{BG} that does not correspond to any IMP. The phase response is shown in Appendix C.

power at the device input P_{in} is calculated by subtracting the input line attenuation from the measurement instrument output power. The input line attenuation was obtained via an ac-Stark shift measurement of a qubit device [19] in a separate experiment.

To probe nonlinearities in the device through intermodulation distortion we use a multi-frequency lock-in amplifier platform [20], where we output two drive tones separated by $\Delta = 100$ Hz, see Fig. 1a). We choose this drive spacing to be significantly smaller than the power-dependent linewidth of the resonator (8-11 kHz). This choice guarantees that both the drives and the lowest-order IMPs remain well within the resonance linewidth.

We measure the response of the device to these two drive tones using a detection frequency comb demodulating the IQ quadratures at frequencies spaced by $\Delta/2$. Consequently, intermodulation produces a signal at every other frequency in this comb, and the remaining detection frequencies sample the noise background. The generation of drive tones and digitization of the response is performed in the second Nyquist zone, without the use of analog mixers for frequency conversion. Odd-order IMPs, i.e. when $|k_1| + |k_2|$ is an odd number, appear as a comb of frequencies when f_c coincides with the resonance frequency f_r , as shown in Fig. 1 b). Even-order IMPs fall far outside of the resonance linewidth.

We perform intermodulation spectroscopy, stepping the detection frequency comb with the two drives in the centre across a frequency range. This frequency range is centred around f_r , with a span well outside the res-

onance linewidth. As shown in Fig. 1 c), we detect no IMPs when the centre of the comb is off-resonant. As the two drives approach f_r , we measure a clear signal at the IMP frequencies, peaking at $f_c = f_r$. The decay of the measured IMPs outside of resonance shows that the IMPs are generated in the device and are not artifacts of nonlinearities present elsewhere in the measurement system. Thus, we can use this intermodulation spectroscopy method to characterize nonlinearities native to our CPW resonator.

III. RESULTS

The standard resonator characterization shows the expected trend where Q_i is suppressed at low $\langle n \rangle$ due to interaction with parasitic TLSs. This observation fits the so-called standard tunnelling model [21]

$$\frac{1}{Q_i} = F\delta_{TLS}^0 \frac{\tanh(hf_r/2k_B T)}{(1 + \langle n \rangle/n_c)^\beta} + \delta_0, \quad (2)$$

where $F\delta_{TLS}^0$ describes the power-dependent loss due to parasitic TLSs, with F quantifying the TLS filling factor, and δ_{TLS}^0 is the dielectric loss tangent, dependent on the density of the TLSs. The parameter n_c is the critical photon number for the saturation of the TLS bath, and the power-independent term δ_0 quantifies the other, non-TLS-related sources of loss, such as radiation or resistive losses due to quasiparticles. T , h , and k_B denote temper-